CHAPTER – 1

INTRODUCTION

As a result of technology's integration into daily life, both personal and societal needs have grown. Over the world, a sizable section of the populace is made up of people who regularly use medication. Given the quantity of prescription medication, there is a significant chance that one may forget to take it at the appointed time or in the recommended dosage. According to data from the American Society of Health System Pharmacists, one-third of senior citizens receive prescriptions for eight or more drugs per day. With so many prescriptions, there is a considerable risk of overlooking taking a pill at the wrong time or, worse, taking the same medication by mistake again.

According to the British Pharmacological Society, dangerous drug responses brought on by dose errors result in hospitalization for over 80% of the elderly. Prescription bottles are covered in stringent instructions, yet people with cognitive disorders like Alzheimer's are not helped by these reminders. The average annual growth rates for the population aged 60 or over are quite high, according to a 2009 United Nations report on ageing and development. Many elderly individuals are single, and others have disabilities that make it more difficult for them to take care of themselves.

Potential health problems could be brought on by any delay or ignorance, such as failing to take medications on time or forgetting to take them at all. It is also challenging for the nurses to be informed on current information about the quantity of pills left and when certain prescriptions need to be refilled. As a result, the desire for something that can raise senior citizens' quality of life is growing. Intelligent gadgets, such as those designed to identify a person's vital signs, notify a medical practitioner in case of an emergency, or identify the origins of specific illnesses after being absorbed by patients, are a few examples.

For elderly patients who must take multiple medications at different times throughout the day, this is especially true. Patients are too indolent to repeatedly adjust alarm clocks, which is a laborious chore. When asked when they should take their medications, a lot of people either fail to provide the exact time or whether they have taken the medication earlier during the day.

1.1. Scope of Work

The Automatic Medicine Intimator, a tool used in this project, is intended to guarantee prompt medication adherence, especially for older patients or people with hectic schedules. Its duties include keeping track of prescription regimens, notifying users visually and audibly, and delivering the right dosage at the right time. Medical professionals or caregivers can remotely monitor the device by integrating it with IoT technologies. A user-friendly interface, battery backup, and warnings for medicine refills are possible extra features. Additionally, by logging compliance data for later analysis, the system can lower the risk of missed or improper medicine doses and enhance healthcare outcomes.

1.2. Problem Statement

People who's having Dementia, are more susceptible to this condition.Improper medicine use can lead to major problems like disease, death, and a delayed recovery. So to remind elderly people, beeper notification is designed. Many people find it difficult to follow prescribed drug schedules because of complex regimens or forgetfulness, particularly the elderly and those with chronic conditions. This may result in health issues, less effective therapy, and higher medical expenses. The absence of caregiver reminders and real-time monitoring exacerbates the issue. An automated system that guarantees accurate dosage, timely medication reminders, and remote monitoring is required to enhance medication adherence and general health outcomes.

1.3 Objectives

The following objectives are:

- To demonstrate an automatic medication intimator system using a PrefBoard for circuit prototyping.
- To conduct a prototype using a DS3231 real time clock module(RTC Module),Push buttons ,LED's, a 16*2 LCD display and a 10K potentiometer.
- To alert the user with LED indicator and a beeper ,when it's time to take their prescription.
- To use visual and audio alerts to make sure that patients are reminded to take their medications on time.
- To automate the precise administration of prescription drug dosages. Minimize the health hazards associated with improper or missed medicine ingestion. Record data on medication adherence for analysis and future use. Offer a portable, dependable, and easy-to-use drug management system.

CHAPTER -2

Literature Survey

1. The research conducted by Nor Saradatul Akmar Zulkifli, Liew Yoon Hong, Mohd Zamri Bin Osman, Virtual Simulation and Computing Faculty of Computing, University Malaysia Pahang Kuantan ©2022 IEEE presents ELA REMIND: A Medicine Box Reminder System for Elderly Living Alone (ELA) implementing the infrared Sensor, Internet of Things and Mobile Application focuses on to help senior citizens who live alone properly manage their prescriptions. It allows family members to remotely check adherence and send timely reminders through the use of an infrared sensor, Internet of Things technology, and a smartphone application. In addition to tracking consumption history and minimizing drug waste, the system notifies users and caregivers when their medications are due. User testing demonstrates how well it works to increase adherence and provide families peace of mind. In order to increase dependability, future updates will support multiple notifications and reminders even while the app is not in use.

2. The Authors Manoj Kumar Shukla,Sarthak Malik, Yerrolla Chanti , School of Electronics and Electrical Engineering,©2023 IEEE proposes Internet of Things based Smart Medicine Box for Monitoring Medications. An IoT-powered smart medicine box is presented in this study with the goal of improving drug management, especially for patients with long-term illnesses. The box automates the distribution and tracking of medications with components such as an STM32 board, IR sensors, a servo motor, and Wi-Fi connectivity. It offers accurate logging, real-time adherence monitoring, and timely reminders via a web or mobile interface. It has been shown to increase adherence and decrease medication mistakes, particularly in older or cognitively impaired individuals. Future improvements could expand its influence on patient care by notifying caregivers of missed medications.

3. The analysis by Mrs. R. Kiruba, Mrs. M. Saranya, Dr. K. Srinivasan, Department of EIE, Sri Ramakrishna Engineering College Coimbatore, India ©2023 IEEE suggests the system Design and Implementation of Internet of Things - based Smart Medicine Box for Patients showcases a smart medication box is introduced to address medication adherence issues, particularly for older patients. The box maintains schedules, automates reminders, and assures correct dosing with its Arduino controller, RTC for timing, and Thing speak for IoT interaction. It alerts users with alarms, LCD screens, and notifications; an infrared detector records drug access and guards against overdosing. Accuracy-tested, the technology improves adherence and gives caregivers real-time updates. In the future, wearable device integration and caregiver notifications might be added for wider uses.

4. The paper by Safwan Sulaiman Sadi, Abdul Awal Tanvir, Arafath Al Hossain, Independent University, Bangladesh Dhaka, Bangladesh,©2024 IEEE. emphasizes IoT-based Real-Time Medicine Reminder and Dispenser for Elderly Patients .This paper describes a Real-Time Medicine Reminder and Dispenser system that uses the Internet of Things to assist senior citizens in remembering to take their medications on time.To guarantee timely dosing, it makes use of smartphone notifications, automatic dispensing,an LCD display, and audible alerts. Through Wi-Fi integration, the solution allows caregivers to monitor and communicate in real time with smart gadgets. It has been proven to be useful in increasing patient autonomy by lowering caregiver dependence and missed doses. For older patients, this creative method enhances medication adherence and overall healthcare results.

5. The system designed by . Karthik,M. Gokul,R. Dhinesh,Department of Electrical and Electronics Engineering,Sri Ramakrishna Engineering College, Coimbatore, ©2024 IEEE incorporates MEDBOT Artificial Medicine Reminder and Deliver Robot with Line Follower Technique using Arduino Uno In Healthcare describes that MEDBOT project combines automated medicine administration and reminder features with a line-following robot to address medication adherence issues for older people and Alzheimer's sufferers. With an Arduino microcontroller with a Real-Time Clock (RTC), LEDs, an LCD display, infrared sensors, and an alarm system, MEDBOT navigates on its own to deliver prescription drugs on schedule. It minimizes errors and lessens reliance on caregivers by

providing visual and aural notifications. This creative approach reduces the burden on healthcare resources, promotes patient autonomy, and improves compliance.

CHAPTER – 3

METHODOLOGY

3.1 Introduction

- Understand target user needs, including the elderly and individuals with mobility impairments.
- Set goals for accurate fall detection, real-time monitoring, and seamless data transmission.
- Design hardware and software systems, integrating sensors, data processing algorithms, and wireless communication.
- Develop a user-friendly interface for data visualization and control.
- Assemble hardware and software components to create an initial working model.
- Conduct testing to refine the system's accuracy and reliability based on performance data and feedback.
- Continue iterating on the prototype to enhance its functionality and address identified issues.
- Prepare detailed documentation for ongoing development and potential future deployment.

3.2 Block Diagram

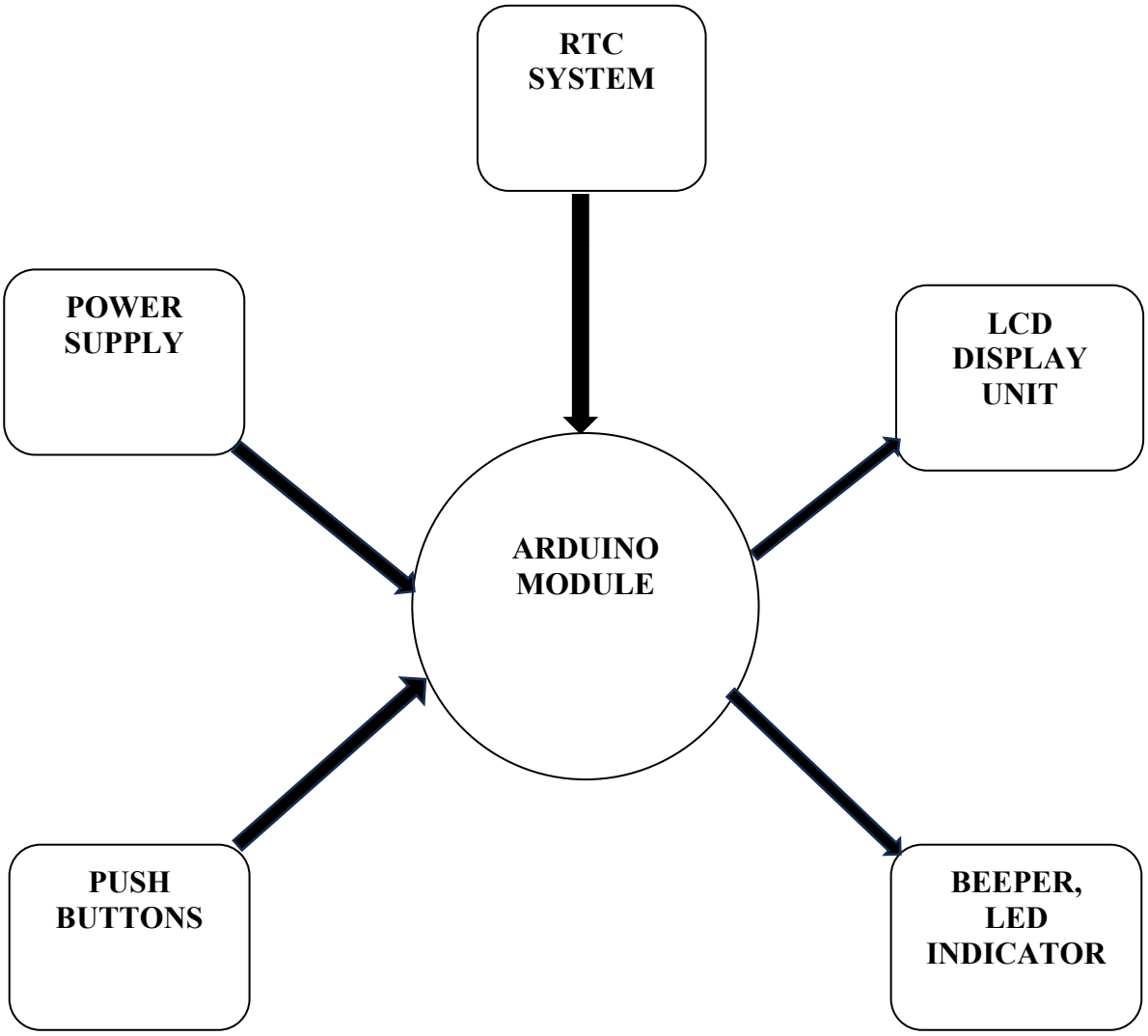

FIG 3.1 Block Diagram

Explanation of Block Diagram:

1.RTC Module: Input : An outside power supply.

Output : Data on the current date and time sent to the Arduino. In charge keeping track of time and transmitting it to the Arduino.

2.Arduino: Inputs : Button inputs for user interaction, time data from the RTC module.

Processing : Updates the display after reading the time from the RTC and comparing it to the preset reminder times.

Outputs: Activates the buzzer for alerts and manages the LCD to show messages.

3.16x2 LCD Display: Input: Arduino data.

Output: Time, reminders, and user prompts are shown visually. gives the user an interface to view the appropriate time to take their prescription.

4.Buzzer: Input: An Arduino alarm signal.

Output: An audible alert to remind you when to take your prescription.

5.Push Buttons: Input: User input to change the time and set reminders. gives the consumer the option to work with the system to personalize their drug schedule.

6.electricity Supply: - Provides electricity to the complete system, guaranteeing that every part works as it should.

Operation:

1. The Arduino receives the time data from the RTC module, which tracks it continually.
2. Using the push buttons, the Arduino compares the current time to preset reminder times.
3. The Arduino notifies the user by updating the LCD display and turning on the buzzer (if applicable) if the current time coincides with a reminder.
4. The push buttons allow users to change settings, which the Arduino then interprets to create new reminders.

This system combines real-time clock tracking, user involvement, and visual/audible messages to make sure users are reminded to take their meds on time. The block diagram makes it simple to comprehend how the system operates by showing how data and control move among the many components.

HARDWARE AND SOFTWARE REQUIREMENTS

3.3. HARDWARE COMPONENTS

1 Arduino Nano
2. 16x2LCD Display
3. DS3231 RTC Module
4. Buzzer
5. LED Indicator
6. Push Button
7. 10K Resister
8. 10K POT
9. Zero PCB

3.4 SOFTWARE REQUIREMENTS

1. The Arduino IDE Software.

1. Arduino Nano

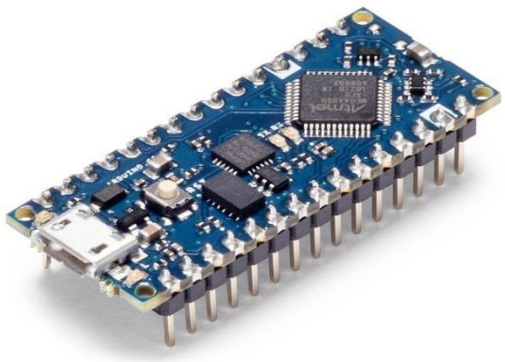

FIG 3.3.1 Arduino Nano

The Arduino Nano is a small, multipurpose microcontroller board that is based on the ATmega328 or ATmega168 microprocessor. It is perfect for applications requiring portability and limited space because of its compact size and USB interface. The Nano has 8 analog inputs, 14 digital input/output pins, and support for SPI, UART, and I2C communication protocols. It can be fueled by an external power supply via a USB, and it runs at a clock speed

of 16MHz. Popular in IoT, robotics, and do-it-yourself electronics, the Arduino Nano is praised for its adaptability, affordability, and ease of use.

2.16x2 LCD DISPLAY

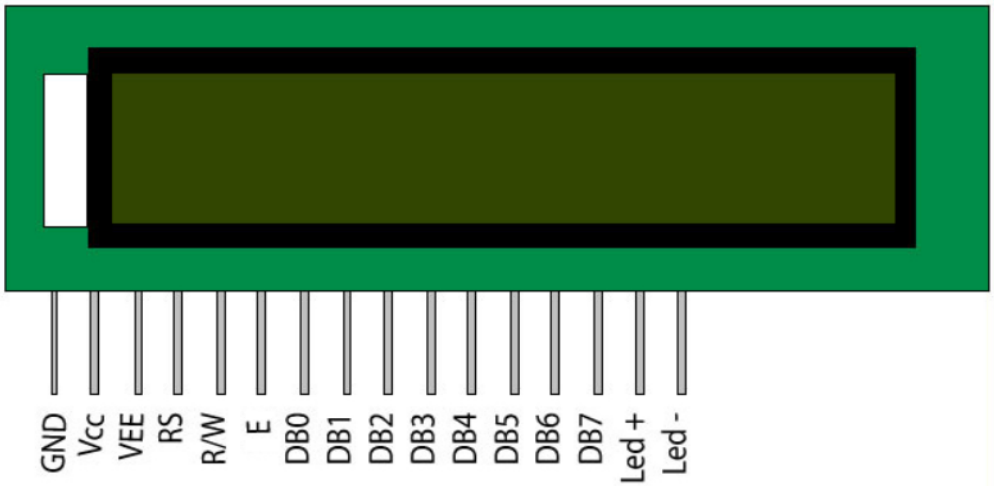

FIG 3.3.2 16x2 LCD Display

In electronics projects, the 16x2 LCD Display module is frequently used to display data. Because it has two rows of 16 characters each, it can produce writing that is brief. It supports basic symbols and alphanumeric characters and is based on the HD44780 controller. Because it uses 4-bit or 8-bit parallel data ports, the display can be used with microcontrollers such as Arduino. It features a backlight for viewing in low light and contrast that may be adjusted. Displaying messages, sensor readings, and system status is a common use for embedded systems.

3. DS3231 RTC Module

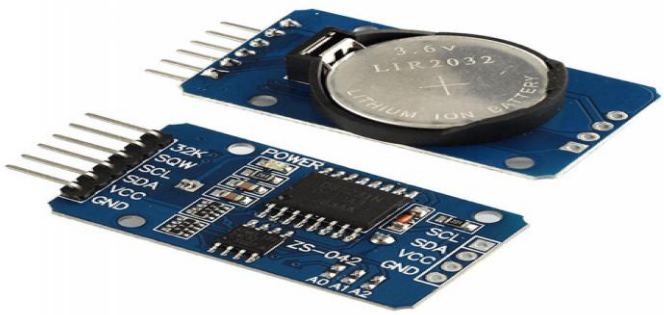

Fig 3.3.3 DS3231 RTC Module

In embedded systems and electronics, the DS3231 RTC (Real-Time Clock) Module is a very precise timepiece. With its built-in temperature sensor and crystal oscillator, it guarantees accurate timekeeping with little drift. The date and time information provided by the module includes the following: day, date, month, year, leap year compensation, seconds, minutes, and hours. In order to function continuously during power outages, it has a backup battery and connects to microcontrollers via an I2C interface. Applications like as clocks, data recording, and scheduling systems are well suited for the DS3231's dependability and simplicity of integration.

4. Buzzer

Fig 3.3.4 Buzzer

A beeper, sometimes referred to as a buzzer, is an electronic gadget that vibrates to emit sound in order to indicate a warning or alarm. It is commonly utilized to provide aural feedback in a variety of applications, including timers, alarms, and electronic gadgets. Both active and passive buzzers are available; active buzzers emit sound when powered on, whilst passive buzzers need an outside input to function. Beepers are small, energy-efficient devices that are frequently employed in consumer electronics, industrial equipment, and embedded systems to efficiently communicate alerts, confirmations, or other audio cues.

5.LED Indicator

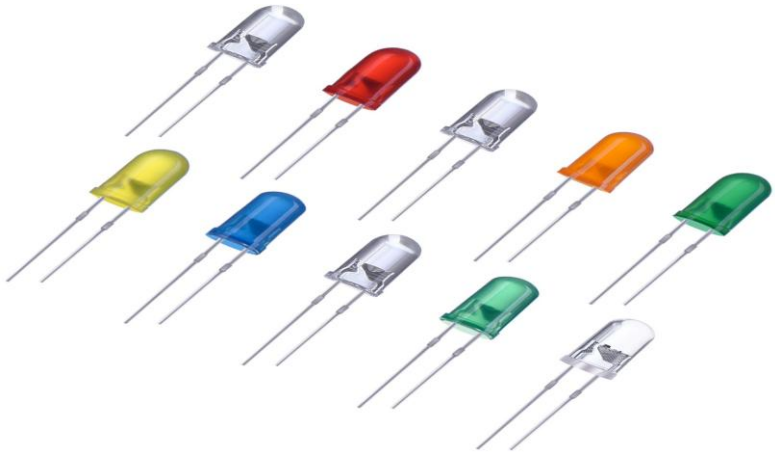

FIG 3.3.5 Led Indicator

LED indicators are small, energy-efficient light sources that are frequently used to indicate the condition of electrical systems or equipment. LEDs are a visible indicator of power status, problems, or specialized tasks and come in a variety of colors. They are perfect for embedded systems, consumer electronics, and industrial applications since they are long-lasting, low power consumption, and robust. They can be used in a variety of products, including control panels, chargers, and alarms. LED indicators are a crucial part of contemporary electrical designs that improve communication and usability.

6. Push Button

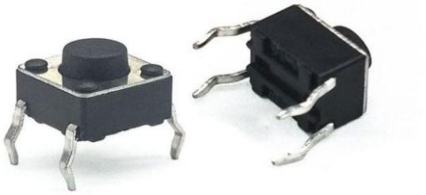

FIG 3.3.6 Push Button

In electronic circuits, a push button is a straightforward, tactile switch that is frequently used to regulate gadget functioning. It completes an electrical circuit when pressed, allowing or initiating a certain activity, and when released, the circuit is opened. Push buttons come in a variety of sizes, shapes, and configurations, including latching and momentary varieties. Because of their dependability and simplicity of use, they are widely utilized in applications such as embedded systems, remote controllers, and user interfaces. They are perfect for electronic projects with limited area because of their small size.

7.10K Resister

FIG 3.3.7 10k Resistor

A 10K resistor is a passive electrical component that has 10,000 ohms of resistance. It divides voltage, limits current, and provides pull-up or pull-down operations for steady signal levels in electronic circuits. It is small, dependable, and adaptable and is usually composed of metal or carbon film. In both analog and digital circuits, the 10K resistor is frequently used in timing circuits, sensor applications, and microcontroller interfaces. Because of its typical resistance value, it is a fundamental part of electronics design and prototyping.

8.10K POT

FIG 3.3.8 10k Pot

In electronic circuits, a 10K potentiometer (POT) is a variable resistor that may change voltage or current. Its resistance ranges from 0 to 10,000 ohms. As it rotates, the resistance changes thanks to a wiper that travels along a resistive track. This enables precise control over variables including brightness, loudness, and sensor calibration. 10K pots are frequently used for user-controlled changes in a variety of applications in embedded systems, audio equipment, and user interfaces. Because of their adaptability and simplicity of integration, they are crucial parts of many electronic projects.

9. Perfboard

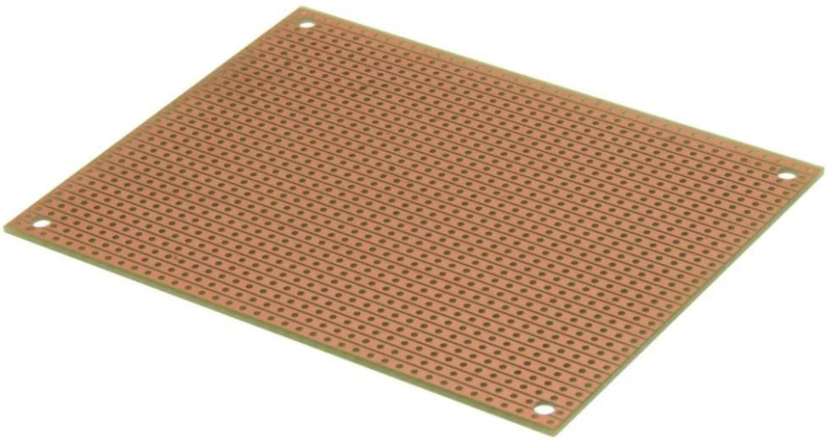

FIG 3.3.9 Perfboard

Perforated boards, often known as perfboards, are flat boards having a grid of pre-drilled holes that are used for electronic circuit assembly and prototyping. It is constructed from fiberglass or phenolic and has copper pads or strips around the holes to ensure that components are soldered firmly. Perfboards are perfect for small-scale projects and testing designs before deciding on a PCB since they enable bespoke circuit layouts. Because they are reusable and long-lasting, engineers, students, and hobbyists utilize them extensively to create effective permanent or semi-permanent electronic prototypes.

1 . Arduino IDE

FIG 3.4.1 Arduino IDE

Arduino IDE Application Logo The Arduino Integrated Development Environment (IDE) is a versatile and user-friendly platform widely used for programming microcontrollers. In this project, the Arduino IDE was utilized to program the Seed Studio Xiao ESP32S3 microcontroller, leveraging its extensive library support and easy-to-use interface. One of the key tasks accomplished using the Arduino IDE was connecting the ESP32S3 to a Wi-Fi network, enabling seamless wireless communication. The Wi-Fi functionality, combined with the Arduino IDE's support for TCP/IP protocols, facilitated the setup of a TCP-Com connection. This connection was crucial for the reliable transfer of motion data captured by the MPU6050 sensor to the Edge Impulse platform for further analysis and classification.

CHAPTER – 4

IMPLEMENTATION

1. Hardware Integration:

In a hardware integration setup, the microcontroller (e.g., Arduino) serves as the brain, controlling and processing data from connected components like the RTC module (DS3231) for accurate timekeeping. The display (OLED/LCD) provides real-time feedback, while the buzzer and LEDs signal alerts and system status. Sensors, such as the infrared sensor, enable object detection or movement, triggering responses from actuators like servo motors for precise actions. A 5V power source, either from a regulator or battery, ensures stable power supply to all components. Optional modules, such as GSM or Wi-Fi, facilitate remote notifications and data transmission. This integration allows for automated actions.

Assembly and Wiring: In the assembly and wiring phase, start by connecting the RTC module (DS3231) to the microcontroller . Attach the LCD or OLED display to the appropriate NANO pins. Wire the buzzer and LEDs to NANO pins, ensuring proper current-limiting resistors are used for the LEDs. For the infrared sensor, connect its output pin to a digital input pin on the microcontroller, allowing it to detect motion or proximity. The servo motor requires a connection to a PWM pin on the microcontroller for precise control over its movement. For power management, isolate the servo motor's power supply from the main system to prevent voltage fluctuations. Use a 5V regulator to supply consistent power to the microcontroller .

Configuration: Set up the microcontroller to control motors and sensors, update the display, and sound warnings in real time. Check connections, adjust parts, and cover the system with an easy-to-use enclosure.

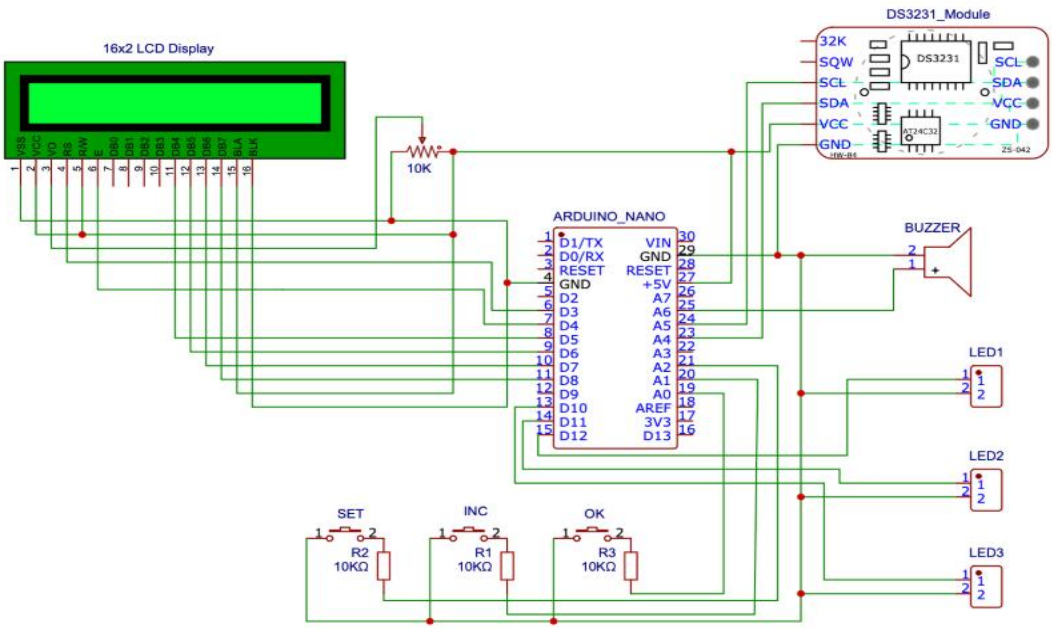

FIG 4.1 Circuit Diagram

Circuit Diagram :

In this circuit diagram, the main part is the Arduino nano microcontroller we used the DS3231 RTC Module and they well connected to the Arduino Pin Numbers

- VCC To 5V
- GND To GND
- SCL To A5
- SDA To A4

When we start the system real-time clock runs in the LCD Display And LCD Will connect to the PIN number

- D3,D4,D5,D6,D7,D8
- The Time we want to set for medication by Using the INC And NEXT buttons Previous Button And Will Connected To The PIN numbers A0, A1, A2
- Led And Buzzer Will Connected To Red Led connected to the Pin Number D10, Yellow Led connected to the Pin Number D11, and Green Led Will connected to The Pin Number D12.also Buzzer Will be connected to the Pin Number A6.

Steps in Implementation:

1.Circuit build: Use the components to create the schematic, then use ZeroPCB to build the layout.

2. Programming: Create Arduino code to take care of managing alarm notifications, keeping track of time, and requesting user input to set reminders.

3. Testing: Check the system's dependability and correctness in reminding users when to take 9 their medications.

4. User Manual: Offer guidance on how to operate the equipment efficiently

2.DataProcessing:

Sensor Data Acquisition :

An RTC module guarantees precise timing for reminders, and the system incorporates sensors such as proximity or infrared sensors to identify compartment access. If weight sensors are used, an ADC is used to convert the analog signals from the sensors into digital data so that medication levels may be tracked. This sensor data is processed by the microcontroller to record events, verify user activities, and detect low drug supplies or missed dosages.

3.Connectivity:

The Arduino IDE is used by the Automatic Medicine Intimator to program the microcontroller. The RTC module, sensors, buzzer, display, and communication modules (GSM/Wi-Fi) are among the components that are connected to the microcontroller via GPIO, I2C, or UART interfaces. Real-time data collection, display updates, and warnings are made possible by the Arduino IDE, which enables the authoring and uploading of code to control these connections. Device communication is made simpler by libraries like `RTClib.h` and `LiquidCrystal.h’, which guarantee smooth system integration and effective operation.

4.ImpulsiveDesign:

There are two primary components to the Automatic Medicine Intimator's ‘Impulse Design’:

1. Processing Block: This block manages the collection of sensor data from gadgets such as weight sensors, proximity sensors, and the RTC module. Real-time inputs are processed, compartment access is detected, and the correct drug retrieval or low supply is checked.

2.Learning Block (Classification): This block classifies user activity patterns using machine learning methods. It can predict reminders, modify timing, and enhance warnings by assessing usage data (such as missed doses and frequency), which increases the system's adaptability to the demands of specific users.

5.Live Classification :

Live Classification : system continuously monitors user interactions through 'real-time data analysis'. The RTC module makes sure that timely reminders are sent, and sensors monitor compartment access. Incoming data is analyzed by the microcontroller, which then categorizes user behavior (such as missing or on-time doses) and modifies subsequent reminders appropriately. The device can identify changes in medicine consumption habits, such as abnormal behavior or missing doses, thanks to 'continuous monitoring'.

Based on this research, the system improves user compliance by offering 'feedback' in the form of alerts, display changes, and optional notifications.

6.TestingandCalibration:

1.Sensor Calibration : To guarantee precise compartment detection, start by calibrating sensors such as proximity or infrared sensors. To avoid erroneous readings, modify the threshold and sensitivity settings.

2.System Testing: Test the system thoroughly to ensure that all of its parts function as a whole. Examine the microcontroller's reaction to sensor inputs, the precision of the display, the operation of the alarm, and the efficiency of the RTC connection modules. To make sure the system sends out proper reminders and notifications, simulate different user scenarios (such as missed dosages and access verification).

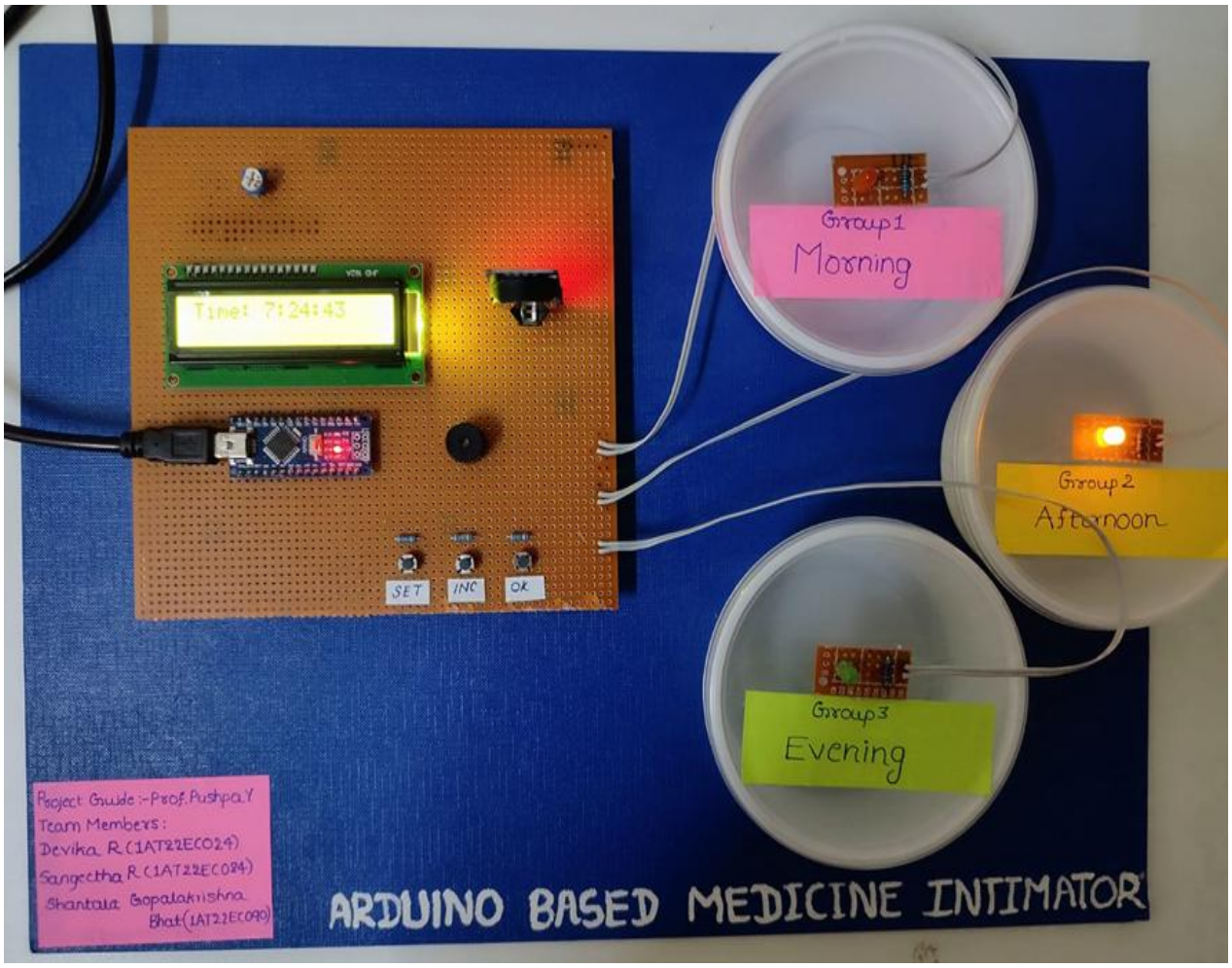

FIG 4.2 Working Model Of Project

Algorithm for the Arduino Based Medicine Intimator System

1. **Initialization:**
    - Begin the communication with RTC, and LCD.
    - Set the pins for input and output:
        - Set the pins as set, inc, ok as input pins.
        - Set buzzer, orange, yellow, and green as output pins.
        - Display a welcome message on the LCD for 3 seconds.
        - Initialize the RTC module. If not running, adjust the RTC with the system time.
2. **Main Loop:**
    - Continuously check the current time from the RTC.
    - Display the current time (hours, minutes, seconds) on the LCD.
    - Call the match() function to check if it's time to remind about any medicine.

3. **Medicine Reminder Setup (Triggered by set_mad button):**
    - When the set_medicine button is pressed:
    - Display a message asking to set reminder times.
    - Prompt the user to input time for three different medicine groups.
    - For each time entry:
        - Use the INC button to increment the hour or minute.
        - Use the next button to confirm the time for each group and store it in EEPROM.

After all three times are set, display a confirmation message.

4. Match Function (Triggered by Current Time):

- For each set reminder time:
- Compare the current time (hour and minute) with the saved reminder times.
- If there's a match:
  - Trigger the buzzer with a beep sound.
  - Show a message on the LCD indicating which medicine group to take.
  - Activate a corresponding LED (orange, yellow, or green).
  - Trigger additional beeps as an alert.

5. Beep Function (Triggering Alert):

- Turn on the buzzer for 500 milliseconds and then turn it off for 500 milliseconds.
- Repeat the beep sound when necessary.

FLOWCHART

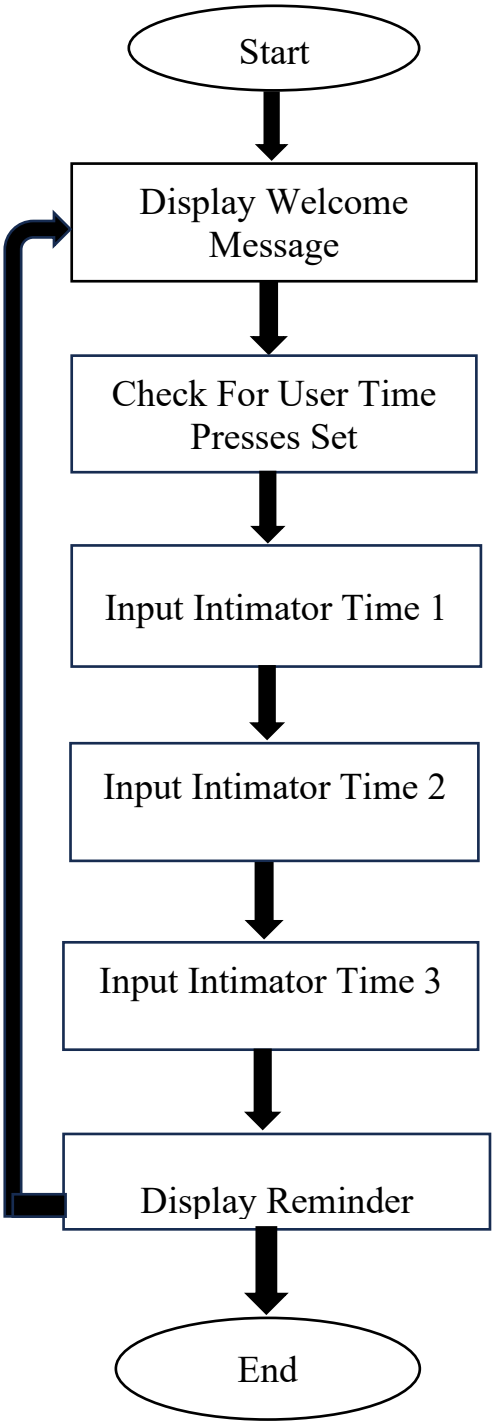

FIG.4.3 Flow Chart

Pseudo Code:

1. Library Imports:

```
#include <Wire.h>        // communication for RTC
#include <EEPROM.h>      // To store reminder times in EEPROM
#include <RTClib.h>      // RTC library for DS1307
#include <LiquidCrystal.h> // LCD library to display messages
```

2. Global Variables:

```
LiquidCrystal lcd(3,4,5,6,7,8);    // LCD pins setup
RTC_DS1307 RTC;                    // Initialize RTC object
int HOUR, MINUT, SECOND;           // Variables to store current time
int hours1, minut, add = 11;       // Variables for storing reminder times
int next = A0, INC = A1, set_mad = A2; // Button inputs for setting reminder times
```

3. Pin Initialization in setup() Function:

```
pinMode(INC, INPUT);  // Button for increasing the time
pinMode(next, INPUT);  // Button to move to the next step
pinMode(set_mad, INPUT);  // Button to start setting medicine reminders
pinMode(buzzer, OUTPUT);  // Buzzer for alarm
pinMode(orange, OUTPUT);  // LED for Group One reminder
pinMode(yellow, OUTPUT);  // LED for Group Two reminder
pinMode(green, OUTPUT);   // LED for Group Three reminder
```

4. RTC Setup:

```
if (!RTC.isrunning()) {
  RTC.adjust(DateTime(_TIME_));  // Set RTC time if not running
}
```

5. Main Loop (loop()):

Displaying the current time:

```
DateTime now = RTC.now();
lcd.setCursor(0, 0);
lcd.print("Time:");
lcd.setCursor(6, 0);
lcd.print(HOUR = now.hour(), DEC);
lcd.print(":");
lcd.print(MINUT = now.minute(), DEC);
lcd.print(":");
lcd.print(SECOND = now.second(), DEC);
```

Handling setting medicine reminder times:

```
if (digitalRead(set_mad) == 0) {
  lcd.clear();
  lcd.print("Set Medicine Reminder");
  delay(2000);
  // Enter time setup for 3 different reminders
  time(1);
  time(2);
  time(3);
}
```

6. Function to Set Reminder Time (time()):

Setting the Hour:

```
if (digitalRead(INC) == 0) {
  HOUR++;  // Increment hour
  if (HOUR == 24) {
    HOUR = 0;  // Wrap around if hour reaches 24
  }
}
```

Saving the Time in EEPROM:

```
if (digitalRead(next) == 0) {
  hours1 = HOUR;
  EEPROM.write(add++, hours1);  // Store hour in EEPROM
}
```

7. Checking if it's Time for the Medicine Reminder (match()):

Compare current time with saved reminder times:

```
int tem[17];  // Array to store times from EEPROM
for (int i = 11; i < 17; i++) {
  tem[i] = EEPROM.read(i);  // Read reminder times from EEPROM
}

if (HOUR == tem[11] && MINUT == tem[12]) {  // Compare with Group One reminder time
  beep();  // Trigger alarm
  lcd.setCursor(0, 0);
  lcd.print("Take Group One Medicine");
  digitalWrite(orange, HIGH);  // Turn on Group One
```

8. Beep Function to Trigger Alarm (beep()):

```
void beep() {
  digitalWrite(buzzer, HIGH);  // Activate buzzer
  delay(500);  // Buzzer on for 500ms
  digitalWrite(buzzer, LOW);   // Deactivate buzzer
  delay(500);  // Buzzer off for 500ms
}
```

Summary of System Behavior :

1. **Initialization:**
   - Configures the pins, LCD, and RTC.
2. **Display Time:**
   - Continuously updates and shows the current time on the LCD.
3. **Set Reminders:**
   - Allows users to set three reminder times via buttons.
   - Stores the times in EEPROM for persistence.
4. **Reminder Alert:**
   - Checks current time against reminder times.
   - If a match is found, the system:
     - Activates the buzzer.
     - Displays a reminder message.
     - Turns on the corresponding LED for the reminder group.

Key Features :

- Persistent Storage: Reminder times remain stored even if the system is powered off.
- User Interaction: Buttons provide an intuitive way to set and confirm reminder times.
- Multi-Group Support: Three distinct reminder groups allow flexibility.
- Audible & Visual Alerts: Buzzer and LEDs ensure the user notices the reminders.

Enhancements (if needed) :

- Add debouncing for button presses to avoid unintended multiple triggers.
- Include functionality to set minutes and seconds for more precise reminders.
- Use a more detailed LCD display to improve user interaction.

CHAPTER – 5

RESULTS

1.Working on the Project is very simple. when we start this system real-time clock runs the time on the LCD.

FIG 5.1 System Overview with Display Interface

2. If we want to set the alarm time for medication we have to press the set_button.

FIG 5.2 Setting The Medicine Intimator Time

3. After pressing this button LCD shows Set Time 1. and then we can select the time we want to set for medication by using the INC Button and the next Button previous process to set the time again. After the Second and Third Time Set the LCD the proper time And Date.

FIG 5.3 Entering The Time

FIG 5.4 Intimator Time Set

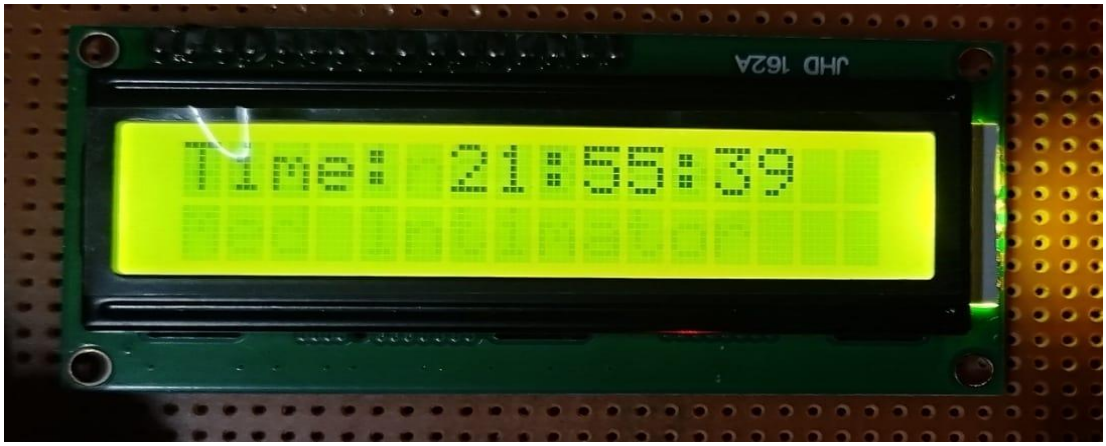

FIG 5.5 Showing Set Time

4. When Any Alarm occurs LCD Indicates Group Medicine 1, Group Medicine 2, Group Medicine 3. The modification Alarm time is also fed in Arduino internal EEPROM to save lost data after a failure.

FIG 5.6 Pointing towards group one medicine

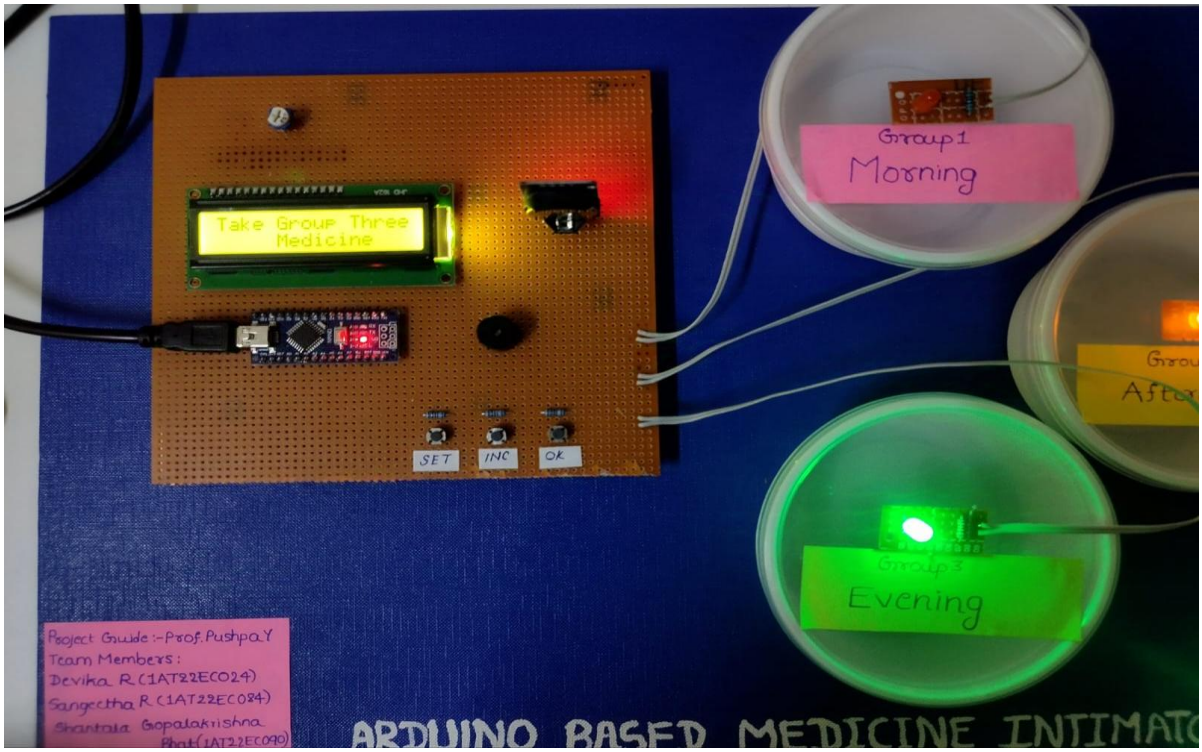

FIG 5.7 Pointing towards group three medicine

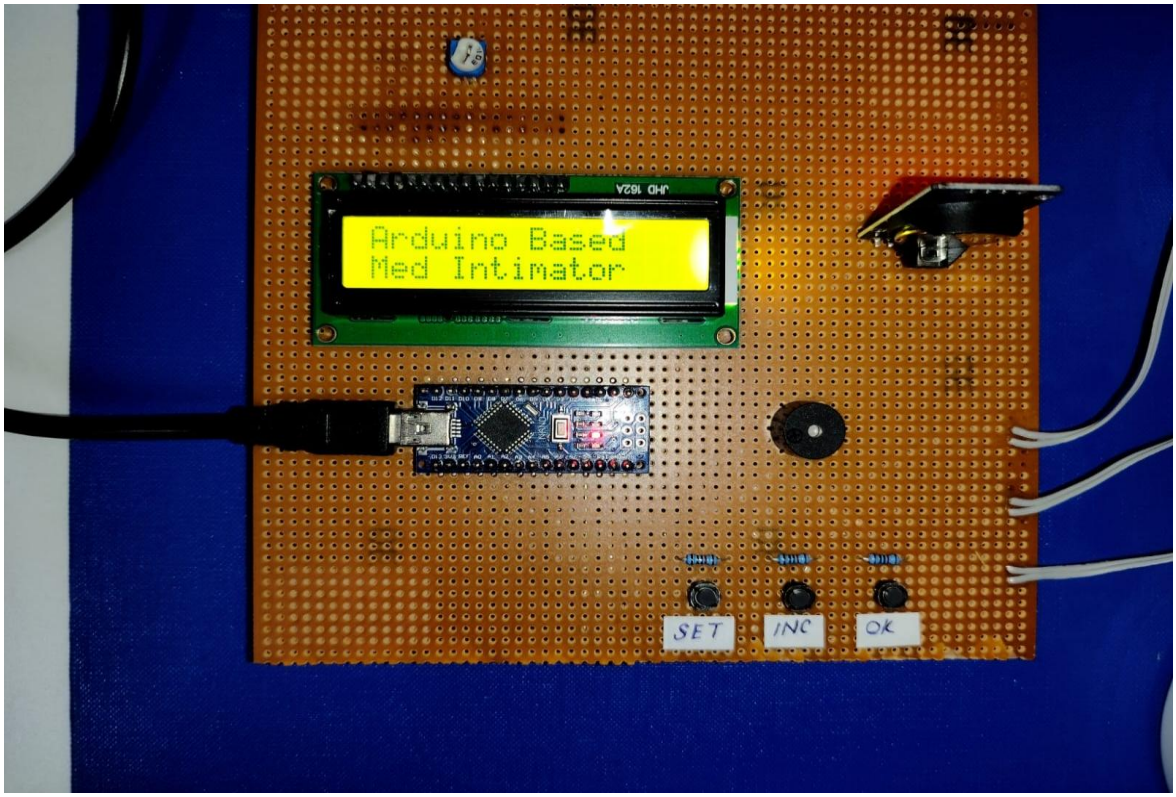

FIG 5.8 Arduino based medicine intimator

Key Outcomes:

1. **Time Display**:
   - Real-time clock (RTC) accurately fetches and displays the current time on the LCD.
2. **Reminder Functionality**:
   - Users can set up to three medicine reminder times using buttons.
3. **Persistent Storage**:
   - Reminder times are stored in the EEPROM and retained even after the system is powered off.
4. **Alarm Notifications**:
   - The system triggers a buzzer and lights up an LED when it's time for a reminder.
5. **User-Friendly Interface**:
   - Simple button controls for incrementing, setting, and saving reminder times.
6. **Group-Based Alerts**:
   - Separate LEDs (orange, yellow, green) indicate reminders for three distinct groups.
7. **Flexibility**:
   - Reminder times can be easily updated or reset as needed.
8. **Reliable Performance**:
   - Ensures timely alerts for medicine intake by matching the current time with stored reminders.

APPLICATIONS AND ADVANTAGES

5.1 Application

1.Older Care: Assures that older people with chronic illnesses receive proper dosage instructions and timely medication reminders.

2.Chronic Illness Management: Assists patients in adhering to daily prescription regimens for ailments such as diabetes, hypertension, and arthritis.

3.Post-Surgery Recovery: Promotes a speedier recovery following surgery by helping to administer prescription antibiotics or medicines on time.

4.Caregiver Support: Gives caregivers the ability to monitor patients remotely and keep tabs on their medication compliance.

5.Use in Clinics and Hospitals: May be utilized in medical institutions to remind patients of their prescription regimens.

6.Mental Health Treatment: Lowers the chance of relapse by assisting individuals with mental health disorders in taking their medications as directed.

7.For homebound patients: who require frequent medication without continual supervision, home healthcare makes medication management easier.

8.Pediatric care: lowers the possibility of mistakes by ensuring that children's drug regimens are taken at the appropriate times and doses.

9.Medication tracking: Provides useful information for medical practitioners by recording medication intake and compliance.

10.Integration with Health Monitoring Systems: This allows for the tracking of vital signs and medication compliance.

5.2 Advantages

1.Better Medication Adherence: Lowers the chance of missed doses by ensuring patients take their medications on schedule.

2.Precise Dosage: Provides the right amount, reducing the possibility of taking too little or too much.

3.Usability: a straightforward and easy-to-use interface for patients, particularly those who are old or have little technical expertise.

4.Remote monitoring: Improves patient care by enabling caregivers or medical professionals to remotely monitor medication adherence.

5.Decreased Error Risk: Medication administration is automated, which reduces the possibility of human error.

6.Customized Alerts: Provides aural and visual reminders based on the patient's choices and schedule.

7.Battery backup: Keeps running in the event of a power interruption to guarantee continuous medicine administration.

8.Data logging: Helps with healthcare management by recording and storing medication intake data for later use.

9.Easy for Busy Schedules: Assists patients with busy schedules in remembering to take their medications without requiring continuous monitoring.

10.Enhanced Independence: Encourages self-care and independence by enabling patients to take their medications with little help.

CHAPTER – 6

CONCLUSION AND FUTURE SCOPE

6.1 CONCLUSION

In summary, those with complex prescription regimens, the elderly, or those with chronic diseases can all benefit greatly from the Automatic Medicine Intimator's ability to improve medication adherence. Error risk is reduced, and overall healthcare management is improved, by giving prompt reminders, precise dosages, and pharmaceutical data logging. Caretakers' remote monitoring capabilities enhance patient care and facilitate better scheduling of treatments. The automatic medicine intimator ensures the safe and efficient administration of medications while supporting independent living with its dependable functionality and easy-to-use design.

6.2 FUTURE SCOPE

The Automatic Medicine Intimator's potential for tailored medicine management rests in its ability to integrate with cutting-edge technologies like machine learning and artificial intelligence (AI). AI could forecast and improve drug regimens by analyzing patient behavior and medical data. Furthermore, incorporating wearable health technology and smart home systems could improve real-time monitoring and automate changes according to health data. These gadgets may give medical practitioners remote, real-time information into patient adherence as telemedicine gains popularity. Additionally, the gadget may develop to accommodate a greater variety of drugs and user requirements, improving patient outcomes and healthcare efficiency even more.

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